On Adversarial Attacks in Geo-Location Prediction Domains

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Abstract

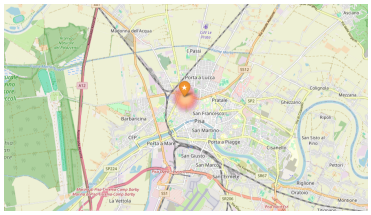
In this work, we will explore different types of attacks and methods in the realm of geodesic predictions [3]. namely we will focus on attacking GeoCLIP [2], a state of the art geo-prediction model. Our project seeks to leverage white-box attack PGDTrimKernel [6] and black-box SparseRS [4], which introduce minimal yet strategically placed patch perturbations in an image to force the model into predicting a significantly different location. We explore both targeted and untargeted attack paradigms. In the untargeted setting, we study how constraints such as the L_0 norm and patch kernel size correlate with the geodesic distance between the true and predicted locations. Conversely, the targeted attack investigates the conditions under which these constraints drive the prediction toward a designated target location. Additionally, we introduce a strategic but small adaptation to SparseRS to match the behavior we observe in PGDTrimKernel, which improves the attack results. Finally, we examine whether perturbations learned on a frozen CLIP model, which is in the heart of GeoCLIP, transfer to GeoCLIP and quantify the resulting impact. The code for the work can be found on GitHub.

1 Introduction

Image geolocation—the process of determining where a photograph was captured—has undergone significant advancement in recent years. Cutting-edge methods, such as GeoCLIP [2], now align images with continuous GPS coordinates rather than relying on discrete cell classifications. GeoCLIP accomplishes this by leveraging a CLIP-based image encoder [7] in tandem with a learned location encoder, thereby mapping images and their corresponding coordinates into a joint embedding space. This continuous representation overcomes the limitations of class-based approaches, which often struggle when an image's true location deviates from predefined class centers. However, the impressive accuracy of these models comes with an inherent trade-off: they pose new privacy risks. Adversaries could potentially exploit geolocation models to infer a user's whereabouts from a seemingly innocuous photo. Prior work has underscored that state-of-the-art vision-language models—even those not specifically trained for geolocation—can inadvertently act as potent location inference tools, raising serious online privacy concerns. In light of these issues, examining adversarial attacks on geolocation systems is essential, both for evaluating model robustness and for exploring defensive measures. In particular, introducing small, carefully crafted perturbations to images may help protect location privacy by misleading these models.



The tower of Pisa in Pisa, Italy



Pisa, Italy

Figure 1: An example of a prediction made by GeoCLIP

In this study, we investigate sparse adversarial patch attacks on the GeoCLIP framework. Unlike traditional L_2 or L_∞ noise-based perturbations, sparse patch attacks alter only a limited set of pixels—governed by an L_0 norm constraint—to simulate realistic threat scenarios (e.g., affixing a sticker or directing a laser at a sensor). The viability of such attacks has been validated in real-world physical settings like in Nemcovsky et al. 2022 [5], reinforcing their practical relevance. By focusing on L_0-constrained modifications, our work models a resourceful adversary who can induce dramatic geolocation errors with minimal alterations to the original image.

Our investigation is further motivated by the central role of the CLIP model [7] in GeoCLIP's architecture. We hypothesize that directly targeting the CLIP encoder can generate adversarial examples that transfer effectively to GeoCLIP, thereby exposing a shared vulnerability. Demonstrating such transferability is particularly valuable because it suggests that perturbations optimized on the more accessible and computationally efficient CLIP model can serve as a proxy for compromising the full geolocation system. In this context, we ask: Can an adversarial patch that disrupts CLIP's image-text alignment also lead GeoCLIP to produce significantly erroneous location predictions, and if so, to what extent?

To do so, we adapt two state-of-the-art sparse patch adversarial attack methods for the geolocation domain. Our white-box approach is derived from PGDTrimKernel [6], modified for continuous space prediction using a margin-based loss to quantify the misalignment in the joint image-location embedding space. In parallel, like PGDTrimKernel, we adapt the query-based SparseRS [4] attack for geolocation, while also enhancing it to allow multiple patch placements instead of only one like in the original version.

In summary, firstly we Adapt PGDTrimKernel and SparseRS Patch Attacks for Geolocation while enhancing SparseRS. Secondly we evaluate the attacks in Targeted and Untargeted Scenarios in investigating the relationship between kernel size and L_0 norm limitation. Lastly we investigate transferability of Adversarial Perturbations from CLIP to GeoCLIP.

2 Related Work

2.1 Terminology

L_0 Norm The L_0 norm of a perturbation δ, denoted as $||\delta||_0$, measures the number of non-zero elements (pixels modified) in the perturbation vector. It quantifies sparsity, thus limiting the total number of altered pixels. Sparse adversarial attacks specifically leverage the L_0 norm

to introduce minimal yet strategic perturbations to deceive models, reflecting more realistic scenarios as these changes can resemble natural variations or subtle manipulations.

Kernel (Patch Kernel) A kernel, or more specifically, a patch kernel in adversarial contexts, defines the spatial dimensions of the adversarial patches applied to an image. Given an image with spatial dimensions (H, W), a kernel defines a smaller rectangular region with dimensions (h, w) where the adversarial perturbation is concentrated.

2.2 GeoCLIP and CLIP

CLIP (Contrastive Language-Image Pre-training) [7] learns shared embeddings between images and textual descriptions by training contrastively on large-scale image-caption datasets. The resulting embeddings generalize well, enabling effective transfer to diverse tasks, including image classification and geo-localization, without task-specific fine-tuning.
GeoCLIP [2] adapts CLIP for geolocation prediction by training a dedicated location encoder alongside CLIP's image encoder. Specifically, GeoCLIP employs continuous random Fourier embeddings for geographic coordinates, hierarchically organizing these embeddings in a coarse-to-fine manner. The image and location embeddings are aligned via a contrastive loss, enabling direct retrieval of coordinates from image queries. GeoCLIP thus surpasses prior cell-based approaches by avoiding discrete partitions, better representing locations sparsely populated in the training distribution. Nevertheless, GeoCLIP inherits CLIP's vulnerabilities, making the examination of CLIP-level attacks meaningful for GeoCLIP's security.

2.3 Adversarial Patch Attacks

Adversarial patches constrain perturbations to small contiguous image regions (patches), representing practical attack scenarios like physical stickers [1, 8]. Unlike widespread noise-based attacks, patches limit perturbations to a region of fixed dimensions, often making them stealthier. Our study utilizes Sparse-RS and PGDTrimKernel due to their state-of-the-art performance. To our knowledge, we present the first examination of adversarial patch attacks against geolocation retrieval models and explore the transferability of patches generated on foundational vision-language models (CLIP) to specialized downstream tasks (GeoCLIP).

2.4 Adversarial Attack Models

Prior adversarial attacks, predominantly investigated in general image classification contexts, offer potential insights for geolocation models. Specifically, sparse attacks—perturbations constrained by the L_0 norm—are realistic scenarios where adversaries modify limited image regions or pixels.

Sparse-RS [4] is a black-box sparse adversarial attack method that leverages iterative randomized search. Given an input image x, a classifier f, and perturbation budget ϵ under L_0 constraints, Sparse-RS finds a perturbation δ that satisfies:

$$\arg\max_{\|\delta\|_0 \leq \epsilon} \mathcal{L}(f(x+\delta), y) \tag{1}$$

where L denotes a loss function. Sparse-RS randomly selects pixel subsets, iteratively explores possible perturbations at these pixel subsets, and retains only the perturbations that im-

prove the adversarial objective. This method efficiently reduces query complexity while achieving high attack success rates even on challenging perturbation patterns like small contiguous patches.

PGDTrimKernel, introduced in SparsePatches [6], employs a white-box gradient-based approach tailored specifically for patch attacks. It begins by optimizing a full-image dense perturbation with standard projected gradient descent (PGD):

$$x^{t+1} = \Pi_\epsilon \left(x^t + \alpha \cdot \text{sign} \left(\nabla_x \mathcal{L}(f(x^t), y) \right) \right) \tag{2}$$

where α is the step size, ϵ is the allowed perturbation magnitude, and Π_ϵ is a projection operator enforcing this constraint. The method iteratively applies a kernel-wise trimming procedure to reduce the perturbation from dense (L_{inf}) to sparse (L_0). Concretely, given a kernel of size $p \times p$, the optimization trims away less impactful kernels based on gradient magnitude, retaining only the highest gradient regions within the L_0 budget. Formally, let M be the binary mask indicating selected pixels:

$$M = \mathbb{I} \left[\text{Top-K} \left(|\nabla_x \mathcal{L}(f(x), y)| \right) \right], \quad \|\delta\|_0 \leq \epsilon \tag{3}$$

where I is the indicator function selecting pixels with gradients in the top-K. The final sparse patch is obtained by applying this mask to the optimized perturbation:

$$x_{adv} = x + M \odot \delta \tag{4}$$

PGDTrimKernel thus jointly optimizes pixel locations and perturbation magnitudes, consistently outperforming previous methods that fixed patch location or sequentially optimized position and perturbation separately.

3 Methodology

In this section, we detail our methodology for performing targeted and untargeted sparse patch adversarial attacks on GeoCLIP. We describe the formalization of the adversarial problem, implementation details of the attacks, and procedures for evaluating the transferability of adversarial patches crafted on the CLIP model.

3.1 Problem Formulation

Consider a geolocation model $f : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^2$, where f maps an image $x \in \mathbb{R}^{H \times W \times C}$ (height H, width W, and channels C) to predicted geographic coordinates (lat, lon). Given a true location y_{true}, the goal of an adversarial attack is to produce a small perturbation δ constrained by an L_0 norm:

$$|\delta|_0 \leq \epsilon \tag{5}$$

such that the perturbed image $x' = x + \delta$ drastically alters the prediction of f.
For an **untargeted attack**, the objective is to maximize the geodesic distance error between the original and adversarial predictions:

$$\arg\max_{|\delta|_0 \leq \epsilon} d(f(x + \delta), y_{true}) \tag{6}$$

where $d(\cdot, \cdot)$ represents the geodesic distance metric.
In contrast, the **targeted attack** aims to minimize the distance between the model's prediction and a specific target location y_{target}:

$$\arg\min_{|\delta|_0 \leq \epsilon} d(f(x+\delta), y_{target}) \tag{7}$$

3.2 Sparse Patch Attack Implementation (PGDTrimKernel)

Table 1: Table with a clean image in the first column, where each other column represents the image being attacked by **PGDTrimKernel** using kernel 1×1, with loosened limitation from left to right on L_0 norm



Clean $\epsilon = 16$ $\epsilon = 128$ $\epsilon = 256$

We implement gradient-based sparse patch adversarial attacks inspired by the PGDTrimKernel method proposed by Nemcovsky et al. [6]. The approach proceeds as follows and uses the code the paper provides with only adjustments for continues tuples classes which represent coordinates on Earth, in targeted and untargeted settings.

3.3 Sparse/Patch-RS Baseline (Modified)

For comparative purposes, we employ an adapted version of Sparse-RS [4], a black-box adversarial attack method with a novel modification. Sparse-RS uses randomized searches over subsets of pixels, iteratively refining perturbations to maximize adversarial loss: 1

Algorithm 1 Adapted Sparse/patch-RS

Input:
$\mathcal{L}$ loss
x image
N max query
k patch area
ϵ perturbation size

$x_{adv} \leftarrow x$
$loops \leftarrow \epsilon \div k$
$q \leftarrow \lceil N \div loops \rceil$
for $i = 1$ to $loops$ **do**
    $x_{adv} \leftarrow$ original Sparse/Patch-RS algorithm with x_{adv}, q
end for
return x_{adv}

Instead of using all N queries to learn a single k-sized square patch perturbation, we split the process into $(\epsilon \div k)$ loops, each running for $q = \lceil N \div (\epsilon \div k) \rceil$ queries. In each loop, we learn a patch of size k, applying the resulting perturbation to the image before proceeding to the next loop. Formally can be seen in 1.



Table 2: Table with a clean image in the first column, where each other column represents the image being attacked by **SparseRS** a kernel size (left to right: $1 \times 1, 2 \times 2, 4 \times 4$) and each row represent the L_0 norm limitation (top down: $\epsilon = 16, \epsilon = 128, \epsilon = 256$)

This adaptation outputs multiple patches similarly to PGDTrimKernel but in a black-box setting. Running the algorithm this way does not hurt performance because of the inherent nature of Sparse-RS; updates to the perturbations occur only when they improve the loss, thus this algorithm is also theoretically sound. In addition, overlapping patches can optionally be disallowed using masks on patched areas, although empirical observations suggest a slight performance trade-off. Finally, we can see that using a kernel the size of the epsilon will output the original algorithm, thus this can be seen more of an extension with more capabilities than a modification to the run of the algorithm.

3.4 Modifying GeoCLIP Architecture

In our implementation, we modified the GeoCLIP architecture to facilitate complete gradient flow throughout the model. By removing gradient blocking operations and restructuring the computational graph, we ensured that gradients could propagate effectively from the output to all trainable parameters. This modification was essential for our PGDTrim kernel white-box attack methodology, as it allowed us to compute and exploit precise gradients with respect to the input.

3.5 Evaluating Transferability from CLIP

To evaluate the vulnerability inherited from CLIP, we craft adversarial patches solely on the CLIP model. Specifically, patches are optimized using CLIP's image encoder and textual prompts representing target locations from the places365 labels (e.g., *"a photo of basketball stadium"*). The adversarial optimization maximizes the similarity between perturbed images and target textual embeddings:

$$\arg\max_{|\delta|_0 \leq \epsilon} \text{sim}(CLIPimg(x+\delta), CLIP_{txt}(\text{target})) \tag{8}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, and $CLIP_{img}$, $CLIP_{txt}$ represent CLIP's image and text encoders, respectively.

The resulting CLIP-crafted patches are subsequently transferred without modification to GeoCLIP inputs to assess the extent to which adversarial features generalize across models. Performance is measured by the induced shift in GeoCLIP's geographic predictions relative to the original unperturbed predictions.

4 Experimental Setup

4.1 Computing Infrastructure

All experiments were conducted on provided GPUs by Technion's Lambda cluster. We implement attacks using PyTorch. GeoCLIP's official implementation [2] was used to load the pre-trained image and location encoders.

4.2 Datasets

We evaluate on standard image geolocation benchmarks. Specifically, we use a sampled subset of Im2GPS3k dataset and a sampled subset of M16 dataset, 75 images sampled using k-means from each. these datasets are widely used for worldwide geo-localization evaluation and traning [2, 3]. While These datasets present a mix of landmark images and ordinary scenes.

We chose them because they are public and used by prior works, allowing us to compare baseline performance. Note that a newer, more uniformly distributed test set called Google-World-Streets-15k (GWS) was introduced by [3], covering 15k Street View images from 193 countries. Unfortunately, this dataset is not publicly available, so we could not include it in our tests – we discuss the implications of this in the Limitations section. Instead, we rely on the diversity of M16 and Im2GPS3k to ensure a broad evaluation. All images are resized to 224×224 for input to GeoCLIP/CLIP.

4.3 Baselines and Metrics

We evaluate two attack methods: our Sparse Patch Attack - PGDTrimKernel [6] and Sparse-RS [4]. Each method is applied in both untargeted and targeted modes. For targeted attacks, we defined only one target, being the location of the Eiffel Tower in Europe $((lat, lon) = (48.858093, 2.294694))$ and for each test image we assign it as the adversarial target. For evaluation metrics, we primarily report the geodesic distance error in kilometers between the predicted and true location (for untargeted attacks, higher is better for the attacker) or between

predicted and target location (for targeted, lower is better for the attacker). We compute median and mean error over the test set. We also report the Attack Success Rate (ASR): for untargeted attacks, ASR is the fraction of images for which the attack causes a prediction at to sway away from the original prediction; for targeted, ASR is the fraction of images where the prediction comes closer to the target compared to the clean prediction.

5 Results

We evaluate the effectiveness of our attacks implementation on the GeoCLIP model under both sparse (individual pixel) and patch-based perturbation regimes. Performance is assessed across five geodesic error thresholds (1km, 25km, 200km, 750km, and 2500km) to capture both minor and severe localization shifts. Our primary focus is the proportion of samples falling within each error threshold, comparing clean predictions against adversarially perturbed ones. Attacks were conducted using different L_0 norm limitations, in both targeted and untargeted settings, using different size kernels (sparse, 2×2, 4×4). See appendix for tables representing attacks with sparse and patch 4×4 kernel attacks.

5.1 Modified Sparse/Patch-RS

Untargeted setting: Across all settings as seen in tables 9.1, 5.1, 9.2, Sparse-RS consistently degraded GeoCLIP's geolocation accuracy, particularly at higher thresholds (750km and 2500km). The clean performance baseline shows that 80% of samples were within 2500km of their true location. Sparse-RS attacks reduced this number to as low as 61.33%—depending on the perturbation size and type. We can observe that sparse attacks (kernel=1×1) are the most effective, showing that the freedom to place pixels independently of each other, and not in batches of kernels, gives us the best results. Although less realistic, it is more effective. as we will discuss in the limitations section, the algorithm was run without any restarts and on a low query budget (1000 queries total per ϵ and kernel size). Because of the nature of Random Search which is in the base of the algorithm, and the selection of the best perturbation each query between the existing one and the previous one, we expect that increasing the budget of the queries and the number of restarts will significantly improve the performance of the algorithm. This is backed up in Croce et al. [4], as the algorithm experiences an exponential increase in performance in the 0-3000 queries budget.

Distance	Clean	L_0 norm limitation						
		$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 16$	$\epsilon = 32$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	7.33%	7.33%	7.33%	7.33%	7.33%	6.67%	6.67%	6.67%
25km	22%	22%	21.33%	21.33%	20%	19.33%	18.67%	**17.33%**
200km	39.33%	38%	37.33%	37.33%	36%	34.67%	34.67%	**31.33%**
750km	60.67%	56.67%	56%	56.67%	53.33%	52%	**52.67%**	53.33%
2500km	80%	72%	70.67%	71.33%	66%	66%	**62.67%**	67.33%
+Changed(%)	–	8	9.33	8.67	14	15.33	19.33	19.33

Table 3: Attack on GeoCLIP with **SparseRS** in an **Untargeted** patch attack setting with **kernel** 2×2. values represent the percentage of examples within a distance from the true location given a L_0 norm limitation (ϵ) ranging from 4 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were pushed away from the true location.

Distance	Clean	L_0 norm limitation						
		$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 16$	$\epsilon = 32$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	0%	0%	0%	0%	0%	0%	0%	0%
25km	0%	0%	0%	0.67%	0%	0%	0%	0%
200km	0%	0%	0%	0.67%	0%	0%	0%	0%
750km	14%	17.33%	17.33%	17.33%	19.33%	20.67%	20%	**21.33%**
2500km	27.33%	30.67%	30.67%	31.33%	32%	34%	31.33%	**34.67%**
+Changed(%)	–	36	35.33	40.67	38	40.67	46.67	**49.33**

Table 4: Attack on CLIP with **SparseRS** in a **Targeted** patch attack setting with **kernel** 2×2. values represent the percentage of examples within a distance from the targeted location given a L_0 norm limitation (ϵ) ranging from 4 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were pushed closer to the targeted location.

Targeted setting: we can see that the model succeeds in moving more than 50% examples towards the target location, which exceeded our expectations given it is double the amount of successful perturbations compared towards Untargeted setting, but on the other hand, the changes are less noticeable within the target radius, meaning while more than half of the attacked examples move towards the goal, most of them don't get within even continent range (2500km). Not shown in tables, but across all perturbations with different variations of $\epsilon and kernel$, the average distance targets are moved is 2000km.

We suspect that the reason why most distance changes that happen between the clean prediction and the adversary one are around 2000km is because of high consecration of GeoCLIP learning in more urban and populous areas as most examples in the dataset will be consecrated in these areas. Thus, if the model predicts a location in Europe, it is highly likely that a perturbation will stir the model to another populous area, which can be in America or China.

On the modification side of the algorithm compared to the original variation, given that all the algorithm modification does is give the option to place more patches given a difference between ϵ and k, it behaves the same exact way as the original when $\epsilon = k$. which means that rather than improving the algorithm performance, it gives it more degrees of flexibility in cases where $\epsilon \geq k$

5.2 PGDTrimKernel

Untargeted setting: Similar to Sparse RS, we've observed a consistent pattern in the effectiveness of adversarial attacks against the GeoClip model. The PGD-based attacks were conducted with a deliberately maintained L_∞ bound of 0.03137 (8/255) to ensure visual imperceptibility, while varying the L_0 sparsity levels from 16 to 256 pixels and kernel sizes from 1 to 4. The model demonstrates strong resilience at low sparsity levels, particularly with L_0=16, where it maintains a robust accuracy of 81.7% averaged across kernel sizes, with the highest robustness of 90% achieved using a kernel size of 4. As sparsity increases, we observe a clear trade-off: at L_0=256, the robust accuracy drops to an average of 70%, with the most significant impact seen at kernel size 1 (65% robust accuracy). The magnitude of prediction displacement also shows interesting patterns, with the largest average displacement of 846.61 km occurring at high sparsity (256) with kernel size 4, while moderate sparsity levels (64) show notably smaller displacements averaging 53.10 km across kernel sizes. This demonstrates that while the model maintains reasonable robustness at very low sparsity levels, it becomes increasingly vulnerable to carefully crafted spatial manipulations as the sparsity budget increases, with the effectiveness of the attacks being further influenced by the choice of kernel size.

Sparsity	**Sparse**	**2X2 Kernel**	**4X4 Kernel**	**Average**
16	164.67	832.39	235.03	410.70
64	49.35	100.24	9.70	53.10
128	325.68	53.78	228.07	202.51
256	218.26	70.31	846.61	378.39
Average	189.49	264.18	329.85	-

Table 5: Average Distance Increase (km) for Different Sparsity and Kernel Size Configurations

Distance	Clean	L_0 norm limitation			
		$\epsilon = 16$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	5%	0%	5%	5%	0%
25km	30%	20%	20%	15%	5%
200km	45%	40%	50%	35%	25%
750km	70%	70%	65%	60%	45%
2500km	90%	90%	80%	70%	70%
+Changed(%)	–	55.00	40.67	46.67	55.00

Table 6: Attack on CLIP with **PGD** in a patch attack setting with **kernel** 4×4. Values represent the percentage of examples within a distance from the original location given a L_0 norm limitation (ϵ) ranging from 16 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were successfully attacked.

5.3 Transfer Learning

Here we attacked CLIP only in an Untargeted setting. We hypnotized that transfer learning will not necessarily translate to better performing attacks, but we also knew that attacks which exploit vulnerabilities in a model, will most likely exploit them in downstream models related to the attacked model, question was, will they lead us closer to the true label or further away from it? being a continues prediction space, fooling the model doesn't necessarily mean we succeed in attacking it to the attackers favor (away from true location / closer to targeted location).
From Tables 9.3, 5.3, 9.3 we can see that the behavior is more sporadic. similarly to previously observed benefits of attacking the base model and transferring the perturbations to downstream models, here the case is more blurry. while we can see a trend that on average, attacking the base model, in our case a specific variation of CLIP, does yield successful attacks on the downstream model, and on average, this attacks significantly change the prediction of the location GeoCLIP provides.

In the tables 9.3, 5.3, 9.3, we decided to include the average overall case (Avg change (km)) which averages the distance gained away from the true location in regards to all the perturbations which were successful on the base model, we can see that on average, no matter what parameters we set in regards to kernel size and L_0 norm limit, we gain 677.60 km on average on all successful adversarial attacks on the base model CLIP after their application on GeoClip. Among these attacks, that brought us closer to the true location (e.g. on average for kernel 2x2 $\epsilon = 16, 256$). Because in this problem domain, just fooling the model does not mean we successfully attacked it. If we discard these perturbations (+Avg change (km) filters out attacks which brought us closer), we sway away from the true location by 3431.28 km on average.

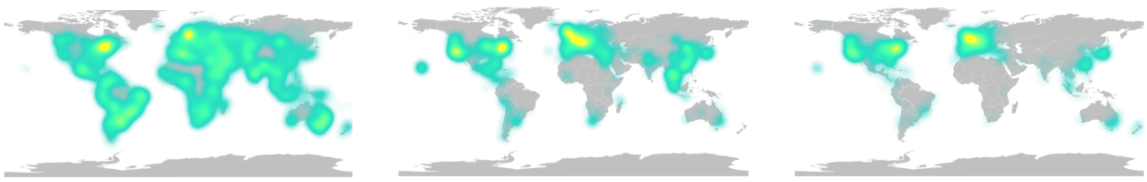
Distance	Clean	L_0 norm limitation						
		$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 16$	$\epsilon = 32$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	6.67%	6.67%	6.67%	6.67%	6.67%	**0%**	6.67%	6.67%
25km	13.33%	6.67%	6.67%	13.33%	6.67%	6.67%	6.67%	6.67%
200km	33.33%	26.67%	26.67%	33.33%	33.33%	33.33%	26.67%	26.67%
750km	60%	60%	60%	66.67%	66.67%	60%	53.33%	53.33%
2500km	66.67%	66.67%	66.67%	73.33%	73.33%	66.67%	**60%**	80%
+Changed (%)	–	20	20	20	26.67	26.67	**46.67**	26.67
+Avg change (km)	–	**4801.97**	4801.97	455.08	3482.77	3431.53	622.67	497.69
Avg change (km)	–	955.09	955.44	-82.71	769.61	901.79	287.20	-77.98

Table 7: Attack on CLIP with **SparseRS** in an **Untargeted** patch attack setting with **kernel** 2×2. values represent the percentage of examples within a distance from the true location given a L_0 norm limitation (ϵ) ranging from 4 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were pushed away from the true location. **+Avg change** represents the average distance examples were pushed away from the true location among only successful example (were pushed further away and not closer). **Avg change** represents the average distance examples were pushed away from the true location among all examples.

6 Limitations

While our study demonstrates effective attacks on GeoCLIP, it has several limitations that are needed to be discussed.

First, computational constraints limited the scope of our experiments. GeoCLIP's inference is computationally heavy. Running iterative attacks on tens of thousands of images was prohibitive with our resources, so we evaluated on 150 samples (Im2GPS3k, M16) rather than the massive global set. This means that our success rates are reflective of those subsets; it is possible that on a truly worldwide random sample (which might include many very challenging images), the attack performance could differ. In addition to that, run times with the GeoCLIPs model were longer than expected, forcing us to use 20 image samples instead of 150 for targeted runs due to time constraints. This limitation also forced us to use less queries, restart tries, and number of epochs for the attack runs.



GWS15k Im2GPS3k YFCC26k

Figure 2: Dataset entries distributed on the map, figure taken from Clark et al. [3]

Secondly, we did not have access to the Google World Streets-15k dataset (GWS15k) introduced by [3]. That data set was designed to cover the globe more uniformly (including many developing regions and countryside images) and to remove biases from previous sets. Without it, our evaluation might still be biased towards regions common in Flickr data (the US, western

Europe, etc.). If GWS15k were available, it would allow testing attacks on a more balanced distribution of scenes. It is possible that in a truly balanced global set, GeoCLIP might be less overfit to certain areas and could either be more robust in general or, conversely, have blind spots we have not seen. We rely on previous analyses from Clark et al. [3] who showed that existing data sets have strong geographic biases. For example, Figure 2 visualizes the distribution of images in YFCC26k and Im2GPS3k: they are 'heavily concentrated in America and Western Europe', with vast regions such as Africa and Central Asia scarcely represented [3]. This imbalance in training data likely affects where the model is confident versus easily confused.

Another limitation is that our adversarial patches are digital. While we drew inspiration from physical patch attacks, we did not test any physical-world scenarios (e.g. pointing a laser at the camera creating the patch distortions). In practice, lighting and angle could reduce the effectiveness of a patch.

Finally, while our transferability experiment is promising, it is limited to one particular backbone (CLIP ViT-L/14 model). GeoCLIP could be extended to other backbones or multi-modal models; the transferability might reduce if the architectures diverge. We also acknowledge that defenses like input pre-processing or robust training on CLIP could potentially mitigate the specific patches we found – we did not test defenses here.

7 Conclusion

In this paper, we presented a first exploration of adversarial patch attacks on GeoCLIP, a state-of-the-art geolocation model leveraging CLIP's image encoder. Our results indicate that sparse patch adversarial attacks can pose a substantial threat to geolocation systems, both in targeted and untargeted scenarios. By adapting white-box and black-box attack methods, we demonstrated the feasibility of driving GeoCLIP's predictions toward large geodesic errors or steering them toward specific target coordinates.

A key observation from our untargeted experiments is the strong sensitivity of GeoCLIP to even a modest number of perturbed pixels when arranged in spatially contiguous patches. Larger L0 budgets typically induced greater displacement from the ground-truth location, though kernel size also modulated the extent of the attack's efficacy. In targeted attacks, we found that more than half of the test images were driven closer to the chosen target—albeit often not within close geographic proximity—highlighting GeoCLIP's susceptibility to targeted manipulations under modest constraints.

We further explored the transferability of adversarial patches from a standard CLIP model to GeoCLIP. Although transfer-induced perturbations did not always move the GeoCLIP prediction away from the true coordinates, they still effected notable shifts in many cases, sometimes on the order of thousands of kilometers. This finding underscores the broader vulnerability inherited from the foundational CLIP encoder: adversarial perturbations optimized strictly for CLIP can partially transfer to downstream geolocation models.

8 Future Work

A prime next step is to evaluate patch-based adversarial attacks on a more uniformly distributed dataset, such as GWS15k, once it becomes publicly available. Such a global-scale test would shed light on the model's resilience (or lack thereof) in regions that are underrepresented in current benchmarks.

Our work focuses on digital patch attacks. Verifying their effectiveness under real-world conditions—such as physical stickers, reflective materials, or directed laser beams—remains an open challenge. These physical adversarial tests would account for factors like lighting variability, camera angle, and real-world occlusions that purely digital experiments cannot fully capture.

Another possible path is to test and see the relations of targeting the base model (CLIP) and see how these gain perturbations will affect the behavior of GeoCLIP. We suspect that aiming the model towards labels that represent less populous areas and places might lead GeoCLIP to mispredict more, although we don't know if it will be closer to the true label or away from it.

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9 Appendix

9.1 Untargeted Patch RS

Distance	**Clean**	L_0 **norm limitation**								
		$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 16$	$\epsilon = 32$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	7.33%	7.33%	7.33%	7.33%	6.67%	6.67%	6.67%	6.67%	6.67%	6.67%
25km	22%	21.33%	22%	21.33%	20%	19.33%	19.33%	16.67%	16.67%	**16%**
200km	39.33%	36.67%	38%	36.67%	36.67%	37.33%	37.33%	33.33%	32.67%	**32%**
750km	60.67%	54.67%	57.33%	54.67%	55.33%	56%	55.33%	51.33%	50.67%	50.67%
2500km	80%	70%	72.67%	70.67%	72%	72%	72%	64%	62%	62%
+Changed(%)	–	10	7.33	9.33	8.67	12.67	12.67	20.67	28	28

Table 8: Attack on GeoCLIP with **SparseRS** in an **Untargeted** sparse setting. values represent the percentage of examples within a distance from the true location given a L_0 norm limitation (ϵ) ranging from 1 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were pushed further than the true location.

Distance	**Clean**	L_0 **norm limitation**				
		$\epsilon = 16$	$\epsilon = 32$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	7.33%	7.33%	7.33%	7.33%	6.67%	6.67%
25km	22%	21.33%	21.33%	20.67%	20%	**19.33%**
200km	39.33%	37.33%	36%	36%	**34.67%**	35.33%
750km	60.67%	56%	55.33%	54.67%	**51.33%**	52.67%
2500km	80%	70%	68.67%	68%	**64.67%**	66%
+Changed(%)	–	10	11.33	12	16	**19.33**

Table 9: Attack on GeoCLIP with **SparseRS** in an **Untargeted** patch attack setting with **kernel** 4×4. values represent the percentage of examples within a distance from the true location given a L_0 norm limitation (ϵ) ranging from 16 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were pushed away from the true location.

9.2 Targeted Patch RS

Distance	Clean	L_0 norm limitation								
		$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 16$	$\epsilon = 32$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
25km	0%	0%	0%	0%	0%	0.67%	0.67%	0.67%	0.67%	0.67%
200km	0%	0%	0%	0%	0%	0.67%	0.67%	0.67%	0.67%	0.67%
750km	14%	17.33%	18%	19.33%	18.67%	18%	18.67%	20.67%	22%	**24%**
2500km	27.33%	31.33%	31.33%	32.67%	31.33%	30.67%	32.67%	32.67%	35.33%	**38%**
+Changed (%)	–	35.33	36	38	38	36	41.33	44.67	53.33	55.33

Table 10: Attack on GeoCLIP with **SparseRS** in a **Targeted** sparse setting. values represent the percentage of examples within a distance from the targeted location given a L_0 norm limitation (ϵ) ranging from 1 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were pushed closer to the target location.

Distance	Clean	L_0 norm limitation				
		$\epsilon = 16$	$\epsilon = 32$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
25km	0.00%	**1.33%**	0.67%	0.67%	0.67%	0.67%
200km	0.00%	1.33%	0.67%	0.67%	0.67%	**2.00%**
750km	14.00%	18.67%	19.33%	18.67%	19.33%	**22.67%**
2500km	27.33%	32.00%	32.00%	33.33%	34.67%	**35.33%**
+Changed(%)	–	50.00	54.00	**57.33**	53.33	56.67

Table 11: Attack on CLIP with **SparseRS** in a **Targeted** patch attack setting with **kernel** 4×4. values represent the percentage of examples within a distance from the targeted location given a L_0 norm limitation (ϵ) ranging from 16 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were pushed closer to the targeted location.

9.3 Untargeted Transfer Patch RS

Distance	Clean	L_0 norm limitation								
		$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 4$	$\epsilon = 8$	$\epsilon = 16$	$\epsilon = 32$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	6.67%	6.67%	6.67%	6.67%	6.67%	6.67%	6.67%	6.67%	6.67%	6.67%
25km	13.33%	13.33%	**6.67%**	13.33%	13.33%	13.33%	20%	13.33%	13.33%	13.33%
200km	33.33%	33.33%	26.67%	26.67%	26.67%	33.33%	33.33%	26.67%	26.67%	26.67%
750km	60%	60%	60%	60%	60%	60%	53.33%	53.33%	**40%**	46.67%
2500km	66.67%	66.67%	66.67%	66.67%	66.67%	73.33%	66.67%	66.67%	**60%**	66.67%
+Changed (%)	–	6.67	13.33	6.67	13.33	20	33.33	40	46.67	**46.67**
+Avg change (km)	–	**13686.7**	359.6	687.53	7187.12	4565.93	3265.95	2843.17	3764.82	2716.18
Avg change (km)	–	907.97	33.28	33.34	943.02	840.4	1011.7	1067.03	**1687.10**	1013.94

Table 12: Attack on CLIP with **SparseRS** in an **Untargeted** sparse setting. values represent the percentage of examples within a distance from the true location given a L_0 norm limitation (ϵ) ranging from 1 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were pushed away from the true location. **+Avg change** represents the average distance examples were pushed away from the true location among only successful example (were pushed further away and not closer). **Avg change** represents the average distance examples were pushed away from the true location among all examples.

Distance	Clean	L_0 norm limitation				
		$\epsilon = 16$	$\epsilon = 32$	$\epsilon = 64$	$\epsilon = 128$	$\epsilon = 256$
1km	6.67%	6.67%	6.67%	6.67%	6.67%	13.33%
25km	13.33%	13.33%	20%	13.33%	**6.67%**	20%
200km	33.33%	26.67%	26.67%	26.67%	33.33%	33.33%
750km	60%	60%	53.33%	53.33%	73.33%	60%
2500km	66.67%	73.33%	66.67%	66.67%	80%	66.67%
+Changed (%)	–	20	20	13.33	13.33	**33.33**
+Avg change (km)	–	4825.99	1108.98	4428.73	330.88	3555.08
Avg change (km)	–	925.69	146.22	551.79	-280.48	1010.74

Table 13: Attack on CLIP with **SparseRS** in an **Untargeted** patch attack setting with **kernel** 4×4. values represent the percentage of examples within a distance from the true location given a L_0 norm limitation (ϵ) ranging from 16 to 256. Clean column represents values given from the attacked model on clean examples. **+Changed** represents the percentage of examples that were pushed away from the true location. **+Avg change** represents the average distance examples were pushed away from the true location among only successful example (were pushed further away and not closer). **Avg change** represents the average distance examples were pushed away from the true location among all examples.